Problem 1.

(a) Let $f : A \to B$, and let C, D be subsets of B. Prove

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D).$$

(b) When E is a countable subset of $\mathbb{R}$, is the complement $\mathbb{R} \setminus E$ always uncountable? Explain why or why not.

(c) When E is an uncountable subset of $\mathbb{R}$, is the complement $\mathbb{R} \setminus E$ always countable? Explain why or why not.

Proof. Start!

(a) Let $f : A \to B$ and $C, D \subset B$. Then

$$\begin{aligned} x \in f^{-1}(C \cap D) &\iff f(x) \in C \cap D \\ &\iff f(x) \in C \wedge f(x) \in D \iff x \in f^{-1}(C) \cap f^{-1}(D). \end{aligned}$$

Thus, $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$, as required.

(b) True! Let E be any countable subset of $\mathbb{R}$. If $\mathbb{R} \setminus E$ is countable, then $\mathbb{R}$ is countable (a contradiction). By the Uncountability of $\mathbb{R}$, $\mathbb{R} \setminus E$ is an uncountable subset of $\mathbb{R}$, as required.

(c) Not True! If $E = [0, 1]$, then $E \supset [0, 1]$ and $\mathbb{R} \setminus E \supset [2, 4]$ are uncountable subsets of $\mathbb{R}$, as required.

□

Problem 2.

(a) State what it means to say that U is *not open*.

(b) Prove that if U is not open, then there exists $x \in U$ and a sequence $\{x_n\}$ of elements of U^c such that

$$\lim_{n\to\infty} x_n = x.$$

(c) Suppose $V \subset \mathbb{R}$ has the following property: for every convergent sequence $\{x_n\}$ of elements of V we have

$$\lim_{n\to\infty} x_n \in V.$$

Prove that V is closed. *Hint*: Argue by contradiction using (b).

Proof. Start!

(a) If $U \subset \mathbb{R}$ and $\exists x \in U \ \forall r > 0 : (x - r, x + r) \cap U^c \neq \emptyset$, we say U is *not open*.

(b) If $U \subset \mathbb{R}$ is not open, then $\exists x \in U$ $\forall n \in \mathbb{N}$ such that

$$x_n \in \left(x - \frac{1}{n}, x + \frac{1}{n}\right) \cap U^c \neq \emptyset \implies |x_n - x| < \frac{1}{n};$$

hence, $\exists$ convergent sequence $\{x_n\}$ of U^c such that

$$\lim_{n\to\infty} x_n = x.$$

(c) Suppose $V \subset \mathbb{R}$ has the following property: $\forall$ convergent sequence $\{x_n\}$ of V,

$$\lim_{n\to\infty} x_n \in V \implies \lim_{n\to\infty} x_n \notin V^c.$$

Assume, for the purpose of contradiction, V is not closed. Then V^c is not open and $\exists$ convergent sequence $\{x_n\}$ of V such that

$$\lim_{n\to\infty} x_n \in V^c \implies \lim_{n\to\infty} x_n \notin V.$$

By contradiction, V is closed.

□

Problem 3.

(a) Use the definition of convergence to prove that

$$\lim_{n\to\infty} \frac{10n^2}{n^2 + 16n + 1} = 10.$$

(b) For each of the following scenarios, give an example satisfying the stated property. Formal proofs are not required, but some explanation may be useful.

(i) A sequence $\{x_n\}$ converging to 0 which is not monotonic.

(ii) An unbounded sequence that has a convergent subsequence.

Proof. Start!

(a) Let $\epsilon > 0$. By the Archimedean Property of $\mathbb{R}$, $\exists M \in \mathbb{N}$ such that

$$M\epsilon > 320 \implies M^2\epsilon > 20.$$

If $n \geq M$, then

$$\left|\frac{10n^2}{n^2 + 16n + 1} - 10\right| = \frac{160n + 10}{n^2 + 16n + 1} < \frac{160}{n} + \frac{10}{n^2} < \frac{160}{M} + \frac{10}{M^2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

(b) Let $x_n \in \mathbb{R}$. By the Archimedean Property of $\mathbb{R}$, $\forall x, y \in \mathbb{R}$ $\exists n \in \mathbb{Z}$ such that $nx \geq y$.

(i) If $x_n = (-1)^n/n \in \mathbb{R}$, then

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} (-1)^n/n = 0$$

and

$$x_1 = -1 < 1/2 = x_2 \ \wedge \ x_8 = 1/8 > -1/9 = x_9.$$

(ii) If

$$x_n = \begin{cases} n & (n \text{ is odd}) \\ 0 & (n \text{ is even}) \end{cases} \in \mathbb{R} \implies \forall x \in \mathbb{R}\ \exists n \in \mathbb{N} : x_n \geq x,$$

then $\{x_n\}$ is an unbounded sequence and $\{x_{2n}\}$ is a convergent subsequence.

□

Problem 4.

(a) Let $\{x_n\}$ and $\{y_n\}$ be bounded sequences of real numbers.

(i) Prove that the sequence $\{x_n + y_n\}$ is bounded.

(ii) Prove that

$$\limsup_{n\to\infty}(x_n + y_n) \leq \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.$$

(b) Let E denote the set of all real numbers in $(0, 1)$ with decimal expansion involving only 1's and 2's:

$$E = \{x \in (0, 1) : \forall j \in \mathbb{N}, \exists d_{-j} \in \{1, 2\} \text{ such that } x = 0.d_{-1}d_{-2}...\}.$$

Note that $0.2 \notin E$ but $0.222222... \in E$. Prove that $0.1111111...$ is a cluster point of E.

Proof. Start!

(a) Let $\{x_n\}$ and $\{y_n\}$ be any bounded sequences of real numbers.

(i) By the Least Upper Bound Property of $\mathbb{R}$,

$$\forall n \in \mathbb{N} : |x_n| \leq \sup_{k\in\mathbb{N}} |x_k| \in \mathbb{R}$$

and

$$\forall n \in \mathbb{N} : |y_n| \leq \sup_{k\in\mathbb{N}} |y_k| \in \mathbb{R}.$$

By the Triangle Inequality of $\mathbb{R}$,

$$\forall n \in \mathbb{N} : |x_n + y_n| \leq |x_n| + |y_n| \leq \sup_{k\in\mathbb{N}} |x_k| + \sup_{k\in\mathbb{N}} |y_k| \in \mathbb{R}.$$

(ii) By the Bolzano-Weierstrass Theorem of $\mathbb{R}$, $\exists$ convergent subsequence $\{x_{n_j} + y_{n_j}\}$ of the sequence $\{x_n + y_n\}$ such that

$$\lim_{j\to\infty}(x_{n_j} + y_{n_j}) = \limsup_{n\to\infty}(x_n + y_n)$$

and $\exists$ convergent subsequence $\{y_{n_{m_j}}\}$ of the sequence $\{y_{n_j}\}$ such that

$$\lim_{j\to\infty} y_{n_{m_j}} = \liminf_{n\to\infty} y_n.$$

Then $\{x_{n_{m_j}} + y_{n_{m_j}}\}$ is a convergent subsequence of the convergent sequence $\{x_{n_j} + y_{n_j}\}$ for which

$$\lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}}) = \lim_{j\to\infty}(x_{n_j} + y_{n_j}).$$

By the Limit Laws of $\mathbb{R}$, $\{x_{n_{m_j}}\}$ is a convergent subsequence of the sequence $\{x_{n_j}\}$ such that

$$\lim_{j\to\infty} x_{n_{m_j}} = \lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}} - y_{n_{m_j}}) = \lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}}) - \lim_{j\to\infty} y_{n_{m_j}}.$$

Thus, $\{x_n + y_n\}$ is a sequence of $\mathbb{R}$ for which

$$\begin{aligned}\limsup_{n\to\infty}(x_n + y_n) &= \lim_{j\to\infty}(x_{n_j} + y_{n_j}) \\ &= \lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}}) \\ &= \lim_{j\to\infty} x_{n_{m_j}} + \lim_{j\to\infty} y_{n_{m_j}} \\ &= \lim_{j\to\infty} x_{n_{m_j}} + \liminf_{n\to\infty} y_n \leq \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.\end{aligned}$$

(b) Let $x_0 = 0.1111111... \in E$ and $x_1 = 0.2222222... \in E$. If $x \in E$, then $\forall j \in \mathbb{N}\ \exists d_{-j} \in \{1, 2\}$ such that $x = 0.d_{-1}d_{-2}...$; hence, $x_0 \leq x \leq x_1$. Thus, $x_0 = \min E$ and $x_1 = \max E$; hence, $x_0 = \inf E$ and $x_1 = \sup E$. If $x_0 < \inf E \setminus \{x_0\}$ and $x_1 > \sup E \setminus \{x_1\}$, then $\exists y, z \in E \setminus \{x_0, x_1\}$ such that $x_0 < y < \inf E \setminus \{x_0\}$ and $x_1 > z > \sup E \setminus \{x_1\}$ (by construction of E); hence, $x_0 = \inf E \setminus \{x_0\}$ and $x_1 = \sup E \setminus \{x_1\}$. By the Supremum and Infimum Characterization of $\mathbb{R}$, x_0 and x_1 are cluster points of E.

□

Problem 5.

(a) Suppose that for all $n \in \mathbb{N}$, $x_n > 0$, $y_n > 0$ and

$$\lim_{n\to\infty} \frac{x_n}{y_n} = L > 0.$$

Prove that $\sum x_n$ converges if and only if $\sum y_n$ converges.

(b) Find all real numbers x such that the power series converges.

(i)
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2020n}(x - 10)^n$$

(ii)
$$\sum_{n=1}^{\infty} n! x^{n!}$$

Proof. Start!

(a) Let $x_n > 0$ and $y_n > 0$ for all $n \in \mathbb{N}$. Assume, for the purpose of direct proof, that

$$\lim_{n\to\infty} \frac{x_n}{y_n} = L > 0.$$

Then $\forall \epsilon > 0\ \exists M \in \mathbb{N}\ \forall n \geq M$,

$$\left|\frac{x_n}{y_n} - L\right| < \epsilon \iff L - \epsilon < \frac{x_n}{y_n} < L + \epsilon \implies (L - \epsilon)y_n < x_n < (L + \epsilon)y_n$$

(given $y_n > 0$ for all $n \in \mathbb{N}$). Consider, for the purpose of direct proof, that

$$\epsilon_0 = \frac{L}{7} > 0$$

(given $L > 0$). Then $\exists M_0 \in \mathbb{N}\ \forall n \geq M_0$,

$$(L - \epsilon_0)y_n < x_n < (L + \epsilon_0)y_n \implies 0 < \frac{6L}{7}y_n < x_n < \frac{8L}{7}y_n < \infty$$

(given $x_n > 0$ for all $n \in \mathbb{N}$). By the Direct Comparison Test, $\sum x_n$ converges if and only if $\sum y_n$ converges, as needed.

(b) We want to find all real numbers x such that the power series converges.

(i) By the Root Test,

$$x \in (9, 11) \iff \lim_{n\to\infty} \sqrt[n]{\left|\frac{(-1)^n}{2020n}(x-10)^n\right|} \left(= \lim_{n\to\infty} \frac{|x-10|}{\sqrt[n]{2020n}} = |x-10|\right) < 1$$

and

$$\sum_{n=1}^{\infty} \left|\frac{(-1)^n}{2020n}(x-10)^n\right| < \infty \implies \sum_{n=1}^{\infty} \frac{(-1)^n}{2020n}(x-10)^n < \infty.$$

If $x = 9$, then

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2020n}(x-10)^n = \sum_{n=1}^{\infty} \frac{1}{2020n} = \infty.$$

If $x = 11$, then

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2020n}(x-10)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{2020n} < \infty.$$

(ii) By the Ratio Test,

$$x \in (-1, 1) \iff \lim_{n\to\infty} \left|\frac{(n+1)!x^{(n+1)!}}{n!x^{n!}}\right| \left(= \lim_{n\to\infty} (n+1)|x|^{n!n} = 0\right) < 1$$

and

$$\sum_{n=1}^{\infty} |n!x^{n!}| < \infty \implies \sum_{n=1}^{\infty} n!x^{n!} < \infty.$$

If $x = -1$, then

$$\sum_{n=1}^{\infty} n!x^{n!} = -1 + \sum_{n=2}^{\infty} n! = \infty.$$

If $x = 1$, then

$$\sum_{n=1}^{\infty} n!x^{n!} = 1 + \sum_{n=2}^{\infty} n! = \infty.$$

□